Observation-anchored event-mechanism editing for fault diagnosis under unseen operating conditions

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ABSTRACT

Operating-condition changes modify the timing and excitation strength of fault-related vibration events, while measured signals also contain structural responses and background components. Enforcing invariance on the entire observation can therefore suppress information useful for diagnosis. This paper proposes observation-anchored event-mechanism editing for domain-generalized fault diagnosis. A source-only sparse analyzer estimates events in a finite response dictionary. The editor changes event locations and effective impulse strength while retaining the response kernels and the remainder of each measured observation. Factual and edited source views train a shared log-spectrum classifier with instance normalization and Mixup. Deployment requires only a vibration window and introduces no additional analysis branch. Experiments cover 27 multi-source tasks on Paderborn bearings, PHM09 gearboxes and JNU bearings, with ten random runs per task. The proposed recipe achieves window accuracies of $83.12 \pm 0.94\%$, $71.98 \pm 0.95\%$ and $99.67 \pm 0.17\%$, respectively. Its dataset-equal accuracy exceeds matched instance normalization and Mixup by 0.80 percentage points and the unedited-bank control by 0.79 points. Same-rate timing-only and global-moment controls distinguish the contribution of selective event editing. On PU, recording-level fault-to-healthy errors decrease from 10.41% to 9.12%; on PHM09, the run-to-run standard deviation decreases from 1.85 to 0.95 points. Comparisons with strong alignment baselines, checkpoint selection and a single-source extension characterize the conditions under which the editor provides useful diagnostic improvements.

1. Introduction

Vibration-based fault diagnosis supports condition monitoring and maintenance by identifying changes in the health state of rotating machinery. Deep diagnostic models can learn discriminative patterns from labeled measurements, but their deployment often involves speeds and loads that differ from those observed during training. A change in operating condition modifies both the excitation of a fault and the way its response appears in a measurement. Consequently, a classifier may fail even when the target fault categories are represented in its training data.

Domain generalization (DG) addresses this setting by training on observed source conditions and evaluating on an unseen target condition. Unlike domain adaptation, it does not require target observations during training. Recent research has developed alignment, contrastive, augmentation and causal learning strategies for this purpose [1, 2]. The application-oriented study of Zhao et al. [1] also shows why empirical risk minimization (ERM), evaluation protocols and the distinction between condition and machine transfer matter when assessing these methods. A useful diagnostic method must demonstrate its increment under the actual information available at deployment.

Causal approaches offer a way to organize this problem around factors that generate the measurements. CIS2N constrains feature dependence and changes between intervention pairs [3]. PC3Net combines prior features, causal-feature learning and collaborative classification for single-source generalization [4]. Anti-causal learning studies the causes of simulated domain shifts [5]. These approaches place useful structure on representation learning. However, operating conditions can affect fault-bearing components themselves: impact spacing, impulse strength and resonance response need not change together. A restriction on an entire representation or waveform does not explicitly distinguish these actions.

This observation motivates a different intervention location. Rather than estimate a condition branch and remove it, we change a specified part of the source observation while retaining the remaining measured signal. Under a convolutional event model, moving an excitation changes its occurrence time without necessarily changing the response kernel. In contrast, dilating an entire waveform moves the excitation and changes the oscillations within its response.

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Moreover, a sparse analysis model cannot be expected to explain every useful component of a real measurement. Replacing the complete signal with its reconstruction would discard that unexplained information.

We propose observation-anchored event-mechanism editing to address these two issues. A numerical source analyzer estimates event coefficients and finite response kernels. The editor moves the estimated events, scales their effective impulse strength according to a fixed rule, and adds the change to the original measurement. The response kernels and the observation-anchored remainder are retained. This construction supplies edited source queries to a conventional diagnostic network; the analyzer is absent from deployment. The method is causal in its motivation and intervention specification within a working signal-generation model. Its learning contribution is evaluated by matched controls that separate event timing, impulse strength, global amplitude statistics and bank sampling.

The main contributions are as follows.

1. An observation-anchored event operator changes event locations and effective impulse strength without dilating the response kernels or replacing the unexplained measured signal. The operator recovers the factual observation exactly at unit rate.
2. A source-only learning procedure integrates the editor with factual supervision and feature-space Mixup. Source analysis is performed offline, and the deployed classifier uses a single vibration window without target speed, a paired recording or an inverse solver.
3. A ten-run evaluation across 27 cross-condition tasks establishes the operator's increment over matched strong controls and examines recording-level errors and repeatability. Timing-only, impulse-only, moment-matched and exposure controls identify which parts of the construction contribute, while a six-direction single-source extension tests its behavior under reduced source coverage.

2. Related work

2.1. Domain generalization for fault diagnosis

DG methods seek a diagnostic rule that transfers beyond the source distributions. Distribution alignment encourages shared feature statistics; domain-adversarial learning discourages source-domain discrimination; contrastive methods use labels to organize inter-domain relationships. CCDG, for example, promotes class-conditioned agreement using a contrastive objective [6]. Deep CORAL and domain-adversarial learning provide established alignment objectives [7, 8]; in a DG setting these objectives can be applied among source domains without supplying target observations.

DG also includes distinct information settings. Multi-source DG learns from several observed conditions, whereas single-source DG must construct useful variation from one condition. PC3Net combines statistical prior features and learned causal features in the latter setting [4]. More recent semi-supervised DG addresses incomplete source labels: the two-stage network of Cui et al. [9] combines pseudo-label construction, feature augmentation and regularized generalization learning. These settings should be distinguished when comparing results. The main study here is supervised multi-source cross-condition diagnosis; the JNU single-source extension is reported separately.

2.2. Causal representations and mechanism-guided construction

CIS2N formulates causal independence and sparse feature shifts, implemented through random Fourier feature covariance and a sparse-shift penalty [3]. MOCD combines sample matching with dual representations intended to distinguish class-related and domain-related information [10]. CFDCR integrates causal representation learning with adversarial adaptation and uses unlabeled target data [11]; it is therefore a related causal approach with a different information setting. These works motivate structural constraints but differ from an explicit edit of an estimated source-signal component.

Physical source construction offers another route. Pang et al. [12] combine analytical fault vibration with measured normal backgrounds. Shi et al. [13] combine semi-physical simulation and contrastive decoupling under speed variation. Physics-guided counterfactual causal learning uses counterfactual regularization and an order-interaction network for single-source cross-machine diagnosis [14]. Thus physical synthesis, measured background and counterfactual learning are established ideas. The present contribution is more specific: each edit retains the remainder of its own labeled observation and changes the estimated event clock while holding the response kernels fixed.

Table 1 summarizes this distinction. The numerical sparse solver [15], instance normalization (IN) [16] and Mixup [17] are adopted components. The proposed method concerns their use with an observation-anchored event intervention, and the experiments isolate this intervention from the benefits of the surrounding classifier recipe.

Table 1
Relationship to representative mechanism-guided approaches. This comparison concerns constructions and information requirements.

Approach	Main learning action	Distinction of the present method
CIS2N [3]	Independence and sparse shifts of learned features	Edits an estimated signal component before feature extraction.
PC3Net [4]	Prior/causal feature collaboration in single-source DG	Uses measured-context event edits and a single deployed classifier.
MOCD [10]	Matched pairs and class/domain representation branches	Keeps the measured remainder without a deployed domain branch.
Fault-model-driven DG [12]	Simulated faults with measured normal backgrounds	Retains the remainder of each labeled observation, including faulty recordings.
PCR-OIPN [14]	Physics-guided counterfactual learning and order interaction	Edits event occurrence with fixed kernels; deployment requires no target speed.
Whole-waveform scaling	Dilation of events, responses and background	Moves estimated events while retaining response shape and observed remainder.

3. Proposed method

3.1. Problem formulation

Let $\mathcal{D}_s = \{(w_i, y_i, d_i)\}_{i=1}^{N_s}$ contain labeled vibration windows from the observed source conditions, where $y_i \in \{1, \dots, K\}$ is a fault label and $d_i \in \{1, \dots, D\}$ identifies a source condition. The objective is to train a classifier that diagnoses an unseen condition using only its vibration window. Target signals, labels, rotational speeds, and target-batch statistics are not inputs to the training algorithm. Recorded source speeds and sampling rates are used by an offline source analyzer. The analyzer and its source partners are not needed at deployment.

The fixed two-rate implementation is denoted Q2. The proposed training construction acts on source observations before feature extraction. It modifies the timing and effective impulse strength of an estimated event component while retaining the response kernels and the unexplained part of the observation. A factual training branch remains in the classification objective throughout training.

Figure 1 summarizes the offline analyzer, source-training path and deployment path. The editor changes the training distribution; it adds no inverse solver or source-matching step to deployment.

3.2. Source event analysis

For a source context $x \in \mathbb{R}^N$, write $\bar{x} = (x - \mu_x)/\sigma_x$, where μ_x and $\sigma_x > 0$ are its saved source mean and scale. The source analyzer fits a nonnegative event model together with harmonic and stationary terms:

$$\bar{x}_j = H_j s_j + A_j b_j + F c_j + \epsilon_j, \qquad s_j \geq 0, \quad j \in \{1, 2\}. \tag{1}$$

Here j indexes two legal source contexts, H_j is a finite convolution dictionary, A_j contains the first four recorded shaft harmonics, and F contains stationary sinusoidal candidates estimated from the source pair. Each response dictionary contains four source spectral carrier candidates, three decay values, and four phases, giving 48 response modes per context. These modes are analysis kernels; they do not mask frequency bands in the classifier input.

The sparse inverse minimizes a squared reconstruction term with nonnegative ℓ_1 penalties on s_j and group penalties on the sinusoidal coefficients:

$$\min_{\substack{s_1, s_2 \geq 0 \\ b_1, b_2, c_1, c_2}} \frac{1}{2} \sum_{j=1}^{2} \|\bar{x}_j - H_j s_j - A_j b_j - F c_j\|_2^2 + \sum_{j=1}^{2} \lambda_{s,j} \|s_j\|_1 + \lambda_b \|B\|_{2,1} + \lambda_c \|C\|_{2,1}. \tag{2}$$

A group in B or C contains the corresponding sine and cosine coefficients across the two source contexts. Fixed proximal iterations produce the offline coefficients and kernels used to construct the source bank using accelerated proximal updates [15]. The classifier does not backpropagate through this numerical inverse.

Source labels are used when selecting same-class source pairs, even though Eq. (2) has no classification loss. The current PU pairing also preserves the recorded bearing identity. A verified entity identity is not assumed for PHM or

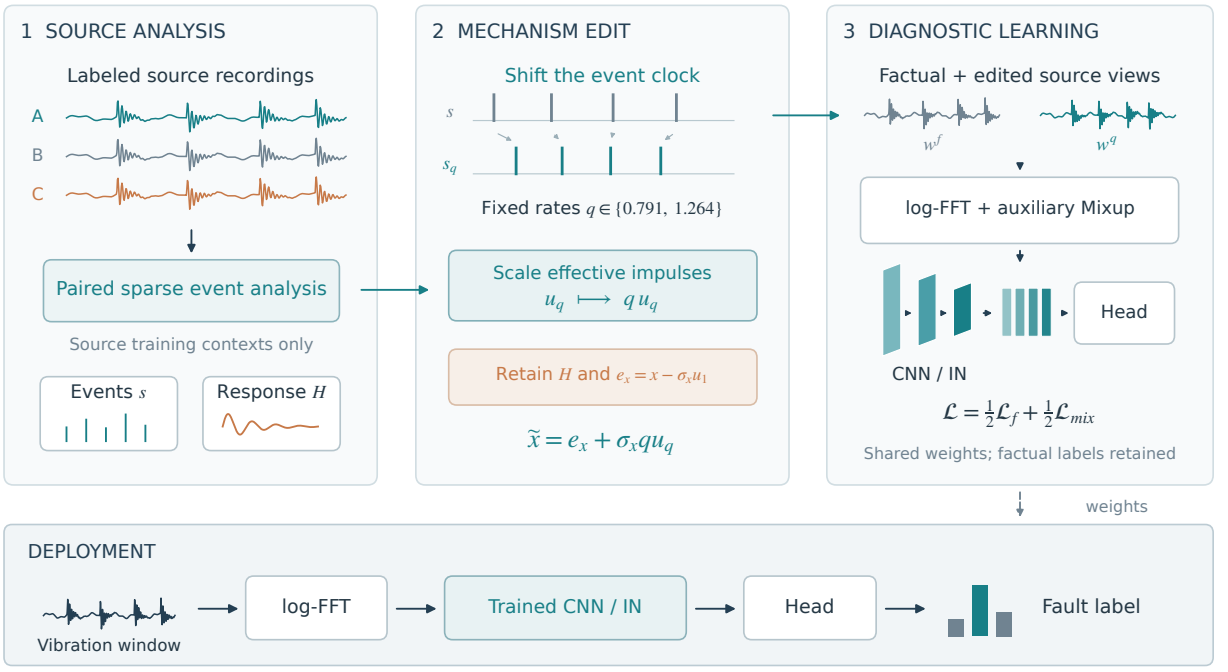


Figure 1: Observation-anchored event-mechanism editing. Source analysis estimates events and response kernels. The intervention changes event timing and effective strength while retaining the response and observation-anchored remainder. Factual and edited views train one shared network; deployment uses the frontend and classifier only. Traces and feature blocks illustrate the processing stages and are not experimental measurements.

JNU. Both contexts must belong to the current source-training role; a source-validation or target context cannot serve as an analysis partner.

3.3. Observation-anchored mechanism editing

For one analyzed context, let $s_b[n]$ denote its event coefficients and h_b its finite response kernels. A kernel is zero outside $[0, L-1]$. Using the context center $m = (N-1)/2$, define a relative event-rate transformation

$$u_q[t] = \sum_{b=1}^{48} \sum_{n=0}^{N-1} s_b[n]\, h_b\left(t - m - \frac{n-m}{q}\right), \qquad q > 0. \tag{3}$$

Thus event locations move from n to $m+(n-m)/q$ while the response shape remains fixed. Fractional lags are evaluated from the same finite analytic kernel used at $q = 1$; they are not created by resampling the complete recorded waveform. The implementation evaluates u_1 and u_q using the same float64 rendering definition and checks context support before extracting a query window.

Define the observation-anchored remainder by

$$e_x = x - \sigma_x u_1. \tag{4}$$

The coupled edit is

$$\begin{aligned} \mathcal{T}_q(x) &= e_x + \sigma_x q u_q \\ &= x + \sigma_x (q u_q - u_1). \end{aligned} \tag{5}$$

The remainder includes the source mean and all components not represented by $\sigma_x u_1$. It is retained rather than assigned to a disposable condition branch. Consequently, the method does not require every condition-dependent feature to be estimated and subtracted.

The relative rate q changes event density and the factor q in Eq. (5) changes effective event impulse strength. The fixed coupling rule scales effective impulses linearly with relative rate. Waveform energy also depends on event overlap, finite support and the retained remainder.

The implementation returns $\mathcal{T}_1(x) = x$ exactly. For other values, the edited query is obtained by cropping the transformed context, $w^q = P_r\mathcal{T}_q(x)$, using the original query location r. No new response or event inverse is fitted for each edit. The current Q2 recipe samples the two fixed values

$$q \in \left\{ \exp\left(-\frac{\log(1.5)}{\sqrt{3}}\right), \exp\left(\frac{\log(1.5)}{\sqrt{3}}\right) \right\} \approx \{0.791286,\ 1.263765\} \tag{6}$$

with equal probability. These factors are the same across datasets and are not chosen from a target rotational speed.

The intervention variables in this event model are event location and effective impulse strength, with the response and remainder held fixed. Its utility is tested by matched timing-only, amplitude-only, global-moment and unedited-bank controls.

The causal motivation is expressed within this event-generation model. Writing $x = \sigma_x\mathcal{H}(s) + e_x$, the action changes the estimated event locations and impulses, while the analyzed response $\mathcal{H}$ and observation term e_x are held fixed. Keeping the training label assumes that the chosen local event changes preserve the diagnostic category. This is a modeling assumption tested through the learning experiments; it does not follow from reconstruction accuracy alone. Effective impulses can vary with operating conditions without changing fault location, but the linear coupling in Eq. (5) is a fixed augmentation rule rather than a universal load–speed law. The construction therefore specifies which mechanism is intervened on without assigning every remaining signal component to a non-causal branch.

The distinction from waveform time scaling follows directly from the kernel model. Set the time origin at the context center and write $\tau_n = n - m$. The event edit and a dilation of the original modeled component then give

$$v_q(t) = \sum_{b,n} s_b[n] h_b\left(t - \frac{\tau_n}{q}\right), \tag{7}$$

$$v_1(qt) = \sum_{b,n} s_b[n] h_b\left(q\left[t - \frac{\tau_n}{q}\right]\right). \tag{8}$$

Both move event centers to τ_n/q. Waveform dilation also scales the time coordinate inside each response kernel. Q2 preserves the response shape and scales effective impulse strength separately by q in Eq. (5). These identities concern the continuous kernel model; subsequent sampling and finite-query cropping produce the training inputs. The harmonic and stationary terms retained in e_x keep their observed coordinates, so q parameterizes an edit of the fitted event clock.

3.4. Classification and feature-space Mixup

A deterministic frontend Φ maps each factual or edited query to $\log(1+|\,\mathrm{rFFT}(w)|)$, followed by linear interpolation to 2048 frequency positions. A shared convolutional encoder, channel-wise instance normalization over the feature position axis, and classification head produce the logits $f_\theta(\Phi(w))$. Raw-valued queries enter this frontend; the context normalization used by the offline analyzer is separate from the classifier input transformation.

Training samples are balanced over source-condition and class cells. For each balanced slot i, a factual window w_i^f is drawn from the full source-training pool and a bank context/query (x_i^b, r_i) is independently drawn from the same condition and class. They need not come from the same recording. Their labels agree because of the balanced slot construction; this is not a requirement to classify one recording using another at test time.

For a binary edit mask M_i, construct the auxiliary feature

$$a_i = \begin{cases} \Phi(P_{r_i}\mathcal{T}_{q_i}(x_i^b)), & M_i = 1, \\ \Phi(w_i^f), & M_i = 0. \end{cases} \tag{9}$$

The mask is sampled with probability $1/2$. At each class and within-cell position it is shared across source conditions, so that edit activation does not introduce a different condition-wise sampling rule. The same sampler and edit-direction streams are used for matched controls.



Figure 2: Analytic illustration of the operator distinction. Columns show the same three synthetic events under the factual model, selective editing and whole-waveform dilation. Rows display events, one response and the resulting observation; dashed traces indicate the original response or retained background. The illustration uses $q = 1.264$ and unit source scale. Q2 moves event centers and scales their impulses while preserving the response and measured-context term. Whole-waveform dilation also changes kernel oscillations and background coordinates. This illustration explains the construction; it is not a measured-signal or classification experiment.

Mixup acts on the auxiliary features after the log-FFT frontend. For a batch permutation π and a shared $\lambda \sim \mathrm{Beta}(0.2, 0.2)$, let

$$\widetilde{a}_i = \lambda a_i + (1 - \lambda) a_{\pi(i)}. \tag{10}$$

The loss combines factual and mixed classification risks:

$$\begin{aligned}\mathcal{L}(\theta) = &\frac{1}{2B} \sum_{i=1}^{B} \ell(f_\theta(\Phi(w_i^f)), y_i) \\ &+ \frac{1}{2B} \sum_{i=1}^{B} \left[\lambda \ell(f_\theta(\widetilde{a}_i), y_i) + (1 - \lambda) \ell(f_\theta(\widetilde{a}_i), y_{\pi(i)})\right],\end{aligned} \tag{11}$$

where ℓ is cross-entropy.

The network forward concatenates three copies of the factual batch and three copies of the same mixed batch. Each of the two distinct input batches therefore occupies $3B$ positions, and the mean risk is Eq. (11). Ordinary batch normalization uses this $6B$ layout during training. All matched operator arms retain the same layout and optimizer schedule. The locked classifier recipe uses Adam for 2000 updates with learning rate 10^{-4}, $(\beta_1, \beta_2) = (0.9, 0.999)$ and zero weight decay.

3.5. Controls and deployment

The matched operator comparison holds the encoder, normalization, sampler, Mixup, optimizer and training duration fixed. The strong simple control uses factual windows for the auxiliary branch. The unedited-bank control

substitutes a clean bank query under the same mask. A timing-only control uses $x + \sigma_x(u_q - u_1)$ at the endpoints $\{2/3, 3/2\}$; another uses the same two interior values in Eq. (6). The coupled Q2 arm uses Eq. (5). The unedited-bank control separates source-bank sampling from event editing, while the same-rate timing-only control tests the additional effective-impulse coupling.

Two additional controls test competing amplitude explanations at the same interior values of q. The amplitude-only edit keeps event locations fixed:

$$\mathcal{A}_q(x) = x + \sigma_x(q-1)u_1. \tag{12}$$

Together with the unedited-bank and same-rate timing-only controls, this gives four matched arms: unedited, timing-only, amplitude-only and coupled. For their scores S_F, S_T, S_A and S_Q, the factorial analysis reports $S_Q - S_A$, $S_Q - S_T$ and the interaction $S_Q - S_T - S_A + S_F$ using the same metric and weighting.

For a clean query $v = P_r x$ and its coupled edit $v_Q = P_r \mathcal{T}_q(x)$, let $\mu(v)$ be the sample mean and $\rho(v) = \sqrt{N_v^{-1} \sum_t (v[t] - \mu(v))^2}$ its centered RMS. The global-moment control is

$$\mathcal{M}_q(v) = \mu(v_Q) + \frac{\rho(v_Q)}{\rho(v)}[v - \mu(v)]. \tag{13}$$

It matches the coupled query's mean and centered RMS before the log-FFT frontend while preserving the standardized waveform shape. When both centered RMS values vanish, the control is the constant $\mu(v_Q)$; a zero clean RMS with a positive edited RMS is rejected rather than replaced with noise. Zero-event and weak-event queries remain in the comparison. All controls retain the same source query, label, mask, Mixup draws and training schedule.

All new source models are frozen before target evaluation. The 2000-step state is the primary evaluation state; the earliest strict best among source-validation steps 0, 1000 and 2000 is a secondary analysis. This reporting policy is not an input to the deployed classifier. At deployment, a vibration window passes through Φ and f_θ in evaluation mode, with source-trained batch-normalization running statistics and per-sample instance normalization. The inverse solver, source partner, q, and rotational-speed metadata are absent from this inference path.

4. Experimental design

4.1. Datasets and tasks

The experiments use the existing PU, PHM09 and JNU recordings. Table 2 summarizes the current protocols. The same tasks and source roles are used throughout the Q2 comparisons. Each task is repeated with seeds 0–9.

Table 2
Current evaluation protocols. Window length and stride are in samples. Source validation is separated by recording for PU and PHM09 and by temporal blocks for JNU.

Dataset	Classes	Length	Stride	Tasks	Transfer setting
PU	3	5120	4096	4	Three conditions to one
PHM09	8	5333	2666	20	Two conditions to one
JNU	4	4000	2000	3	Two speeds to one

PU uses the Paderborn bearing benchmark [18]. The four recorded conditions are A: 1500 rpm, 0.1 Nm, 1000 N; B: 1500 rpm, 0.7 Nm, 400 N; C: 1500 rpm, 0.7 Nm, 1000 N; and D: 900 rpm, 0.7 Nm, 1000 N. The current inventory contains 18 bearing IDs, each recorded 20 times per condition at 64 kHz. Labels are the retained healthy, inner-race and outer-race coarse categories. For each bearing, 16 recordings per source condition are assigned to training and four to validation using the recorded seed-specific split. The first 60 complete windows of each recording are used; their last sample boundary is 246784, so padding is unnecessary. Bearing identities repeat across conditions. This protocol evaluates transfer to an unseen condition within the recorded bearing collection. It is different from the historical six-domain PU protocol that also combines damage-collection membership.

PHM09 uses the gearbox challenge data [19]. Its ten domains combine shaft rates 30, 35, 40, 45 and 50 Hz with High and Low loads: A/B, C/D, E/F, G/H and I/J, respectively. Sampling frequency is 66666.67 Hz. The eight labels

denote the retained gearbox configurations, with two recording files per configuration and condition. There are ten adjacent-domain and ten distant-domain tasks; their exact source–target pairs are listed in Appendix A. Recording-grouped source splits retain source-training class coverage. Configuration labels are diagnostic categories, not gearbox serial identities.

JNU [20] has healthy, inner-race, outer-race and rolling-element labels at 600, 800 and 1000 rpm, recorded at 50 kHz. A, B and C denote these speeds. Each source recording is divided into non-overlapping training and validation time blocks; window intervals do not cross the split. All three leave-one-speed tasks are retained. The source recordings are not independent new-bearing replications. The three leave-one-speed tasks already exhaust the two-source/one-target combinations of these three speeds. JNU is retained as an additional cross-dataset applicability case; the non-saturated PU and PHM09 tasks provide the main discrimination among strong recipes. The existing three-dataset summary is retained, with per-dataset results reported alongside it. A separately reported single-source extension evaluates all six cross-speed directions using source-local numerical analysis (Appendix C.1).

All contexts and paired observations used by the event analyzer lie entirely inside the corresponding source-training roles. PU pairing retains recorded bearing identity; PHM09 and JNU use same-class legal source pairs without inventing an entity ID. Three centered query windows are taken from each 32768-sample source context for the intervention bank. Target data are used only after all new source models in a comparison are frozen.

4.2. Training and comparison design

The matched operator study uses the same convolutional encoder, feature normalization, classification head and balanced source sampler. The factual batch size is $B = DK\lfloor 120/(DK)\rfloor$, yielding 117, 112 and 120 for PU, PHM09 and JNU, respectively. The actual forward uses the $6B$ layout described in Section 3. All matched arms use Adam with learning rate 10^{-4} for 2000 updates and the same initialization and random-draw streams within each task and seed. Source validation is evaluated at 0, 1000 and 2000; the final state is primary and earliest strict source best is secondary. The encoder has three 1D convolutions with 8, 16 and 32 output channels and kernel lengths 33, 17 and 3. Each convolution is followed by batch normalization, LeakyReLU with slope 0.1 and average pooling of width 2. Feature instance normalization, with variance denominator equal to the feature length and $\epsilon = 10^{-5}$, is applied after the final convolutional block. Flattening gives an 8192-dimensional vector. Fully connected layers of widths 256, 256 and 128 precede the K-class output, with LeakyReLU after each hidden layer. This network is shared by all matched controls. Table 3 gives the preparation and learning procedure.

Table 3
Source preparation, learning and deployment procedure.

1. Partition the observed source recordings into training and validation roles. Select same-class training-context pairs using the declared source metadata; retain bearing identity for PU.
2. Form the response, shaft-harmonic and stationary dictionaries. Run the fixed numerical inverse in Eq. (2) for each registered source pair.
3. For each source context, render u_1 and the two fixed-rate u_q signals. Construct $\mathcal{T}_q(x)$ with Eq. (5), then crop the three observation-supported queries and cache their log-FFT features.
4. For each training update, sample a balanced factual batch and same-class, same-source-condition bank slots. Draw the shared edit mask and rate direction; form the auxiliary features using Eq. (9).
5. Draw the Mixup coefficient and permutation. Update the shared classifier using Eq. (11) with the common $6B$ forward layout.
6. Freeze the classifier after 2000 updates for the primary evaluation. Evaluate source-validation selection separately as a secondary policy.
7. At deployment, apply the log-FFT frontend and trained classifier to each vibration window. Average probabilities only when reporting recording-level diagnosis.

4.3. Baseline implementations and comparison scope

The comparisons address two questions: the performance of complete diagnostic recipes and the contribution of the proposed editor. The complete-recipe panel includes an ERM classifier using the shared CNN and IN, source-domain DANN and Deep CORAL, author-backbone CE and IN+Mixup, adapted CCDG and two documented MOCD interpretations. The ERM, DANN and Deep CORAL baselines use the same source pools, class/domain-balanced sampling, 2000 updates and Adam learning rate 10^{-4}; they use cross-entropy without Mixup or event editing.

DANN applies a source-domain discriminator through gradient reversal. Deep CORAL averages pairwise covariance alignment penalties over source domains. Neither receives target samples.

The author-backbone CCDG family retains its 3001-update schedule and the author's feature-layer contrastive core, with declared input length, class-count and data-role adaptations. A separate MOCD family uses its two-stage 400+400-update schedule. Its equation-literal and prose/stop-gradient implementations remain separate entries; the prose interpretation is included in the representative main table and both are retained in the complete appendix. These rows compare implemented complete recipes under the current tasks, rather than reproducing the original papers' headline scores.

Operator attribution uses the matched controls of Section 3. They share the CNN, IN, Mixup, optimizer, batch layout, steps and source pools with Q2. The unedited-bank arm also shares the source query construction. The proposed Q2 recipe is kept fixed across all three datasets. A half-exposure comparison changes only how often the already-defined edit is applied; it does not define a new primary method. The entire multi-source panel contains 22 recipes. PCR is evaluated on aligned PU recording roles, and the single-source extension is evaluated separately on JNU.

4.4. Metrics, repetition and checkpoint selection

Window accuracy is the fraction of correctly classified windows; window macro-F1 averages class F1 over the declared label set. Recording predictions are obtained by averaging window probabilities within each recording and then taking the maximum-probability class. Recording balanced accuracy (BA) averages class recalls. Window accuracy weights windows within a task, whereas recording metrics give each recording one prediction. These two evaluation units are kept separate.

For dataset d, task set $\mathcal{T}_d$ and run r, the dataset score is

$$S_{d,r} = \frac{1}{|\mathcal{T}_d|} \sum_{t \in \mathcal{T}_d} S_{t,r}, \qquad S_r = \frac{1}{3} \sum_{d=1}^{3} S_{d,r}. \tag{14}$$

Tables report the mean and sample standard deviation of the ten run scores, with denominator nine for the variance. The dataset-equal summary S_r gives PU, PHM09 and JNU equal weight; a task-equal summary over 27 tasks is also retained in the accompanying results. Run seeds affect initialization and sampling and, where specified by the protocol, source train/validation assignment. The reported dispersion therefore describes repeated execution of the complete experimental recipe.

The primary checkpoint is the fixed final training state. For Q2 and matched controls this is update 2000. The secondary policy selects the earliest strict best source-validation state from updates 0, 1000 and 2000. For matched multi-source arms, the selection score averages window accuracy equally over source domains; the JNU single-source extension uses source class-balanced accuracy. Each baseline uses its corresponding declared checkpoints. Model selection does not use target scores, and all source models in each comparison are frozen before target evaluation. The repeated-run results use the fixed tasks and recipes; the dataset panel had already been examined during method development.

For paired interpretation, the ten differences between Q2 and a control are formed before computing their mean and descriptive 95% t_9 interval. These intervals are post-result, unadjusted summaries of variation across runs on the fixed panel. They do not treat overlapping windows or repeated tasks as independent physical experiments. The main performance evidence is the complete mean±SD tables, per-task differences and matched component controls.

5. Results and analysis

5.1. Cross-condition diagnosis performance

Table 4 shows the fixed-final-state results. Q2 attains 83.12%, 71.98% and 99.67% window accuracy on PU, PHM09 and JNU, respectively. Relative to matched IN+Mixup, the increments are 1.68, 0.69 and 0.03 percentage points. The dataset-equal mean is 84.92%, with an increment of 0.80 points; nine of the ten paired run averages are positive. The corresponding increment over the unedited-bank control is 0.79 points, also positive in nine runs. These comparisons retain the surrounding learning recipe and provide direct evidence for the added source intervention.

The strong conventional baselines qualify the overall ranking. ERM reaches 82.97% on PU and 99.71% on JNU, while its PHM09 accuracy is 63.81%. Deep CORAL reaches 82.99%, 72.65% and 99.87%, respectively. Q2 is higher on PU but lower on the other two datasets; its dataset-equal window accuracy is 0.25 points below Deep CORAL.

Table 4
Cross-condition window accuracy (%) at the fixed final state. Entries are mean ± sample standard deviation over ten runs.

Method	PU	PHM09	JNU
Author-backbone CE	80.63 ± 1.38	51.37 ± 1.93	83.95 ± 7.88
Author-backbone IN+Mixup	81.85 ± 0.85	68.23 ± 1.06	95.31 ± 4.86
CCDG (author feature core)	81.13 ± 1.16	57.47 ± 1.73	87.19 ± 8.11
MOCD (prose interpretation)	80.50 ± 0.93	65.26 ± 2.51	95.14 ± 2.86
ERM (shared CNN+IN)	82.97 ± 1.34	63.81 ± 0.99	99.71 ± 0.14
DANN (source domains)	80.81 ± 1.52	52.94 ± 2.40	82.47 ± 10.02
Deep CORAL (source domains)	82.99 ± 1.14	72.65 ± 1.25	99.87 ± 0.06
Matched IN+Mixup	81.44 ± 0.98	71.30 ± 1.85	99.64 ± 0.08
Unedited source bank	81.47 ± 1.09	71.29 ± 1.90	99.64 ± 0.13
Proposed Q2	83.12 ± 0.94	71.98 ± 0.95	99.67 ± 0.17

Complete recipes share the evaluation roles but retain the stated backbone and schedule adaptations. The last three rows share the complete Q2 training recipe; their difference isolates bank sampling and editing.

Table 5
Window macro-F1 and recording balanced accuracy (BA), in percent, over ten runs.

Metric	Method	PU	PHM09	JNU
Macro-F1	ERM	81.51 ± 1.34	58.51 ± 1.21	99.67 ± 0.17
Macro-F1	Deep CORAL	81.27 ± 1.00	68.69 ± 1.44	99.84 ± 0.07
Macro-F1	Matched IN+Mixup	80.07 ± 1.02	67.75 ± 2.15	99.62 ± 0.09
Macro-F1	Proposed Q2	81.57 ± 0.85	68.66 ± 1.18	99.65 ± 0.17
Recording BA	ERM	86.97 ± 2.43	66.16 ± 1.02	100.00 ± 0.00
Recording BA	Deep CORAL	86.86 ± 1.92	75.56 ± 1.52	100.00 ± 0.00
Recording BA	Matched IN+Mixup	86.58 ± 1.24	74.94 ± 2.95	100.00 ± 0.00
Recording BA	Proposed Q2	87.81 ± 0.83	76.72 ± 1.71	100.00 ± 0.00

The performance of DANN and the author-backbone recipes varies more substantially across datasets. Thus, complete-recipe differences are useful for assessing competitiveness, while the matched comparisons locate the event editor's contribution.

Table 5 reports complementary metrics. Q2 improves window macro-F1 over matched IN+Mixup by 1.50 points on PU and 0.91 points on PHM09. Recording BA increases from 86.58% to 87.81% on PU and from 74.94% to 76.72% on PHM09. Q2 also has higher recording BA than Deep CORAL on these two datasets, although Deep CORAL has higher PHM09 window accuracy. This difference shows why a single aggregate accuracy does not fully describe diagnostic behavior. On JNU, the strong recipes are close to the ceiling, and recording BA is 100%; this case primarily demonstrates compatibility with another dataset.

5.2. Performance across individual target tasks

Figure 3 locates the aggregate changes. On PU, the largest benefit occurs when the target is condition D, at 900 rpm: Q2 improves over matched IN+Mixup from 38.43% to 44.84%. The other three targets are already above 91%, and their changes are much smaller. Deep CORAL remains stronger on D at 46.93%. Q2's PU advantage is consequently concentrated in a difficult speed-transfer task, together with smaller gains on two other targets.

PHM09 contains both favorable and unfavorable transfers. Q2 gains 10.33 points over matched IN+Mixup on AB→G and 7.24 points on IJ→D, but loses 8.83 points on IJ→B. Gains therefore do not follow only from the number of source conditions or from calling a task extrapolative. The full task table in Appendix C retains these differences. The three JNU multi-source tasks have little remaining headroom; their single-source counterparts provide a more demanding extension without redefining the main protocol.

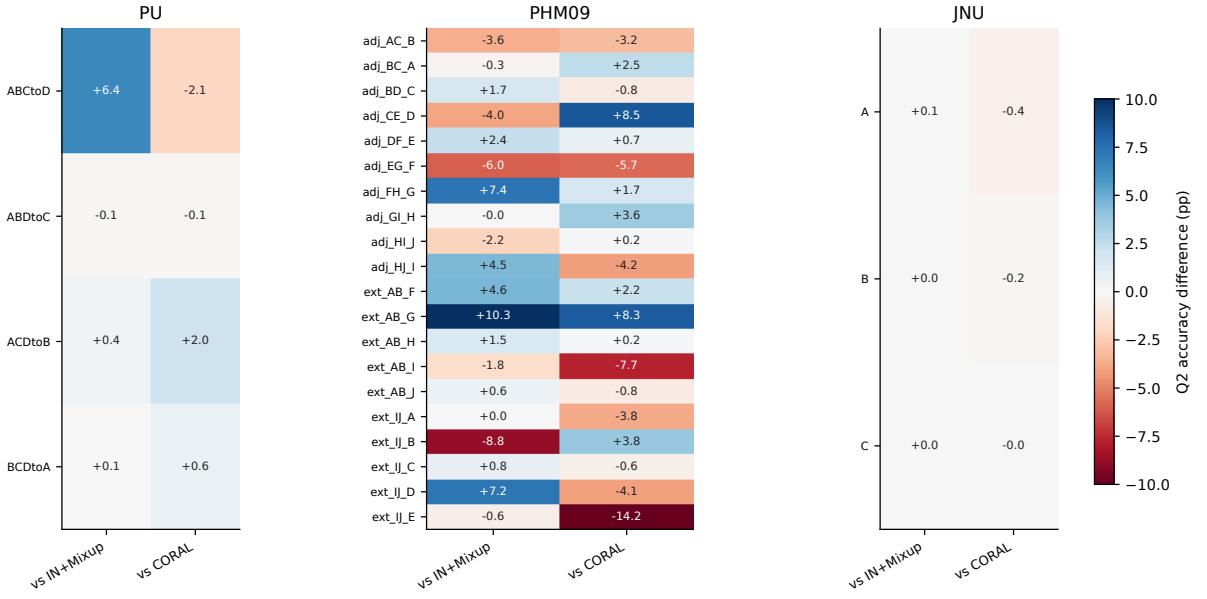


Figure 3: Ten-run mean window-accuracy differences on every multi-source task. Blue denotes a Q2 gain and red a loss. The color scale is clipped at ±10 points; cell annotations retain the actual values. All tasks are shown, including unfavorable target conditions.

Table 6
Matched operator ablations: window accuracy (%), ten runs, fixed final state.

Source construction	PU	PHM09	JNU
Matched IN+Mixup	81.44 ± 0.98	71.30 ± 1.85	99.64 ± 0.08
Unedited bank	81.47 ± 1.09	71.29 ± 1.90	99.64 ± 0.13
Timing, endpoint rates	82.82 ± 1.02	71.59 ± 1.16	99.62 ± 0.19
Timing, Q2 rates	82.84 ± 0.85	71.26 ± 0.84	99.63 ± 0.13
Impulse strength only	81.46 ± 1.19	72.58 ± 1.52	99.66 ± 0.12
Global moment matching	81.36 ± 0.87	71.69 ± 1.27	99.69 ± 0.07
Proposed Q2	83.12 ± 0.94	71.98 ± 0.95	99.67 ± 0.17

5.3. What contributes to the event editor?

Table 6 distinguishes bank sampling from editing and separates the two intervention components. The unedited bank gives 81.47% on PU and 71.29% on PHM09, close to matched IN+Mixup. Q2 improves these means to 83.12% and 71.98%. Consequently, drawing auxiliary queries from a prepared source bank does not by itself account for the observed increment.

At the same Q2 rates, timing-only editing gives 82.84%, 71.26% and 99.63%. Adding the effective-impulse rule increases dataset-equal accuracy by 0.346 points, with a positive difference in all ten runs. Conversely, adding timing to impulse-only editing increases the dataset-equal mean by 0.355 points, although impulse-only editing is stronger on PHM09. The whole-query moment control matches the edited query's mean and centered RMS, yet remains 0.676 points below Q2 in the dataset-equal summary. The useful change is therefore not fully explained by a global amplitude adjustment.

The two intervention components are not uniformly synergistic. The accuracy interaction $S_Q - S_T - S_A + S_F$ is +0.280 points on PU, −0.569 on PHM09 and +0.020 on JNU. The combined construction is beneficial in the main aggregate, but these interactions do not establish a positive coupling effect on every dataset. The endpoint timing recipe is also competitive: its dataset-equal mean is only 0.247 points below Q2. The evidence favors the specified local event construction over the matched simple recipe, rather than an unrestricted claim that stronger event changes always improve diagnosis.

5.4. Repeatability and checkpoint policy

Table 7
Run-to-run dispersion of window accuracy (percentage points). The dataset statistic is the SD of ten task-averaged scores; within-task SD is averaged over tasks.

Dataset	Dataset SD		Mean within-task SD		Lower-SD tasks
	IN+Mixup	Q2	IN+Mixup	Q2	Q2 / total
PU	0.976	0.943	1.572	1.403	3/4
PHM09	1.848	0.950	5.234	4.669	13/20
JNU	0.079	0.167	0.144	0.208	1/3

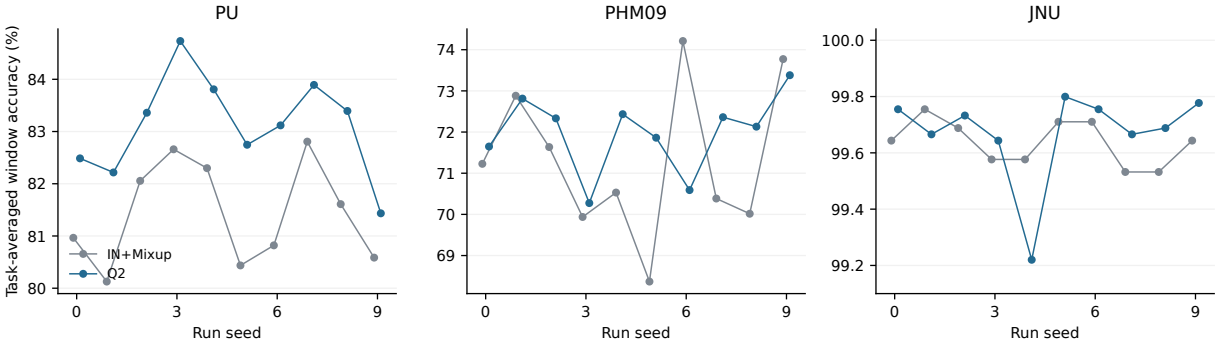


Figure 4: Dataset scores for all ten runs under the fixed-final-state policy. Each point first averages the tasks within its dataset. The vertical scales differ to resolve variation near the JNU ceiling.

The PHM09 improvement is accompanied by lower run-to-run dispersion. Its dataset-level accuracy SD falls from 1.848 to 0.950 points, a 48.6% reduction. The average within-task SD also decreases, from 5.234 to 4.669 points, and 13 of 20 tasks show a reduction. This agreement between aggregate and task-wise summaries supports improved repeatability on PHM09. PU exhibits a smaller aggregate reduction, from 0.976 to 0.943 points, with lower SD on three of four tasks. JNU does not show this pattern: its aggregate SD rises from 0.079 to 0.167 points. Repeatability is therefore a dataset-dependent benefit, not a general noise-robustness claim.

Table 8
Source-validation-selected window accuracy (%) over ten runs. This is a secondary checkpoint policy.

Method	PU	PHM09	JNU
Deep CORAL	83.04 ± 1.04	71.97 ± 1.14	99.88 ± 0.06
Matched IN+Mixup	81.38 ± 0.92	71.40 ± 1.50	99.72 ± 0.12
Unedited bank	81.49 ± 1.16	71.39 ± 1.46	99.71 ± 0.11
Timing, endpoint rates	82.68 ± 1.01	71.79 ± 1.07	99.70 ± 0.10
Timing, Q2 rates	82.82 ± 0.86	70.92 ± 0.93	99.69 ± 0.12
Impulse strength only	81.53 ± 1.08	72.95 ± 0.92	99.69 ± 0.09
Global moment matching	81.59 ± 1.35	73.00 ± 1.55	99.75 ± 0.09
Proposed Q2	83.00 ± 1.05	71.58 ± 1.22	99.74 ± 0.06

Table 8 evaluates the source-selected policy on the same trained trajectories. Q2 gives 83.00%, 71.58% and 99.74%, and its dataset-equal advantage over matched IN+Mixup is 0.61 points. This is positive in eight of ten runs. On PHM09, the global-moment control reaches 73.00% and Deep CORAL reaches 71.97%, both above Q2. The primary and secondary policies therefore support related but distinct conclusions; the best target outcome is not used to choose between them.

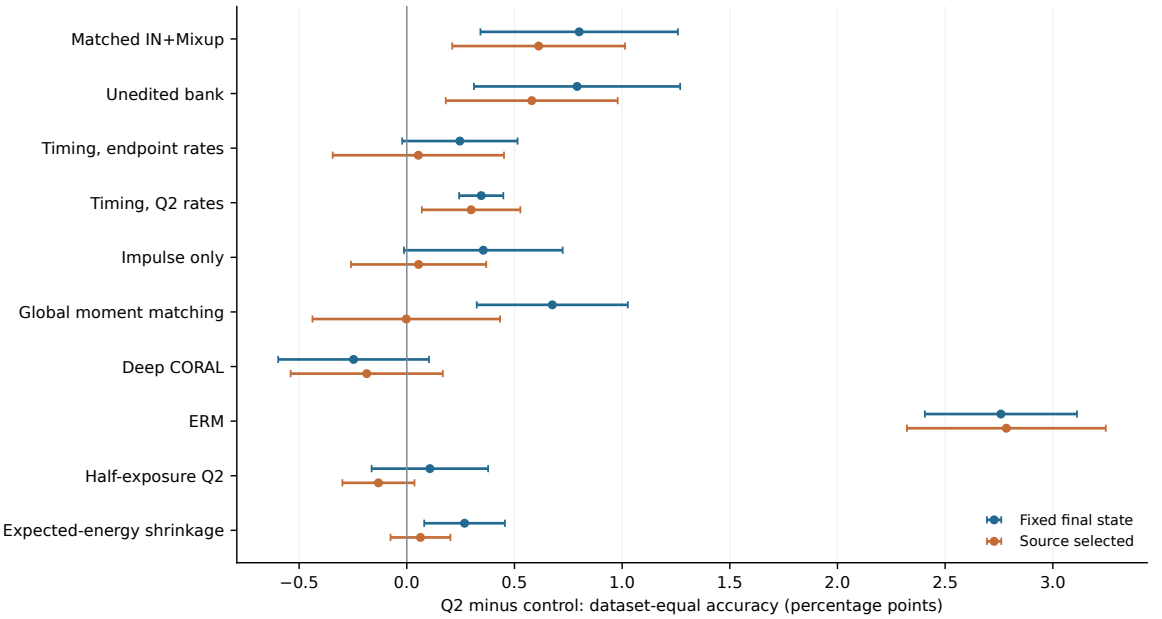


Figure 5: Paired dataset-equal accuracy differences under both checkpoint policies. Bars are descriptive 95% t_9 intervals over ten paired run averages, without multiple-comparison adjustment. Zero indicates the control mean. These intervals summarize run variation conditional on the fixed task panel.

Figure 5 gives the paired uncertainty summaries. For the primary policy, the Q2-minus-IN+Mixup interval is $[0.342, 1.259]$ points; the same-rate timing-only comparison is $[0.243, 0.448]$. The comparison with Deep CORAL is $[-0.597, 0.103]$. These descriptive results complement, rather than replace, the dataset and task tables. Complete paired values are reported in Appendix B.

5.5. Recording-level diagnostic errors

Table 9
PU recording-level error rates (%), ten runs. A lower value is better.

Method	Healthy false alarms	Faults called healthy	Inner/outer confusion
Matched IN+Mixup	3.69 ± 2.37	10.41 ± 1.79	7.88 ± 0.84
Unedited bank	3.69 ± 2.25	10.38 ± 1.38	7.41 ± 0.88
Proposed Q2	3.19 ± 1.52	9.12 ± 1.06	7.57 ± 0.71
Deep CORAL (source domains)	6.62 ± 4.78	8.43 ± 1.65	7.97 ± 1.72

Healthy false alarms use healthy recordings as denominator. The other two rates use all faulty recordings. Scores are computed per task and then averaged within each run; they are empirical classification errors, not system failure probabilities.

Table 9 translates the PU classification results into two practically different errors. Relative to matched IN+Mixup, the fraction of faulty recordings called healthy decreases from 10.41% to 9.12%, while healthy false alarms decrease from 3.69% to 3.19%. Confusion between inner- and outer-race faults changes from 7.88% to 7.57%. These are ten-run, task-averaged recording rates calculated from the saved confusion matrices. They show an improvement in fault recognition without an accompanying increase in the mean false-alarm rate.

Deep CORAL has a lower fault-to-healthy miss rate of 8.43%, but a higher healthy false-alarm rate of 6.63%. Q2 and Deep CORAL therefore make different diagnostic trade-offs. No operating threshold is tuned on target data, and the reported errors use the original maximum-probability decisions. The measurements concern classification performance and are not estimates of machine failure probability or maintenance cost.

5.6. Comparison with physics-guided counterfactual learning

Table 10
PU recording-level comparison, ten runs, fixed final state. All scores are percentages.

Recipe	Accuracy	Macro-F1	Balanced accuracy
Q2, short vibration input	84.81 ± 1.04	84.00 ± 1.08	87.81 ± 0.83
Matched IN+Mixup	83.33 ± 1.39	82.74 ± 1.49	86.58 ± 1.24
Unedited bank	83.78 ± 1.24	83.03 ± 1.15	86.92 ± 1.13
Native 40-rev. author CE	85.58 ± 1.65	82.07 ± 1.37	84.59 ± 1.96
Native 40-rev. CE+real healthy	85.32 ± 1.95	82.35 ± 1.72	85.68 ± 2.06
Native 40-rev. PCR	82.51 ± 0.64	80.28 ± 0.97	84.37 ± 1.09
Short-input author CE	84.30 ± 0.84	76.39 ± 0.67	75.79 ± 0.94
Short-input CE+real healthy	84.09 ± 0.93	76.10 ± 0.93	75.91 ± 0.94
Short-input PCR	83.99 ± 0.98	75.95 ± 1.14	74.99 ± 1.61

Native PCR uses recorded speed to construct 40-revolution inputs. The comparison aligns recording roles, not input windows or architectures. This is a declared cross-condition adaptation of a single-source cross-machine method.

Table 10 compares Q2 with the implemented PCR-OIPN variants [14]. Their native 40-revolution inputs cover different windows from the short-input models. Predictions are therefore compared after averaging probabilities over the same PU recording roles. Q2 gives 84.81% recording accuracy, 84.00% macro-F1 and 87.81% BA, compared with 82.51%, 80.28% and 84.37% for native PCR. The native author CE baseline attains higher recording accuracy at 85.58%, but lower macro-F1 and BA. The comparison does not imply that all benefits arise from counterfactual learning or that Q2 dominates all evaluation metrics.

The information requirements also differ. The native angular frontend uses recorded target speed to form the 40-revolution input; Q2's deployed frontend uses vibration alone. All variants retain their declared frontends and training rules. This experiment evaluates a multi-source cross-condition adaptation, whereas the original PCR study addresses single-source cross-machine diagnosis. It is included as a mechanism-related comparison, without substituting published results obtained under a different protocol.

5.7. Reduced source coverage on JNU

Table 11
JNU single-source extension: all six source–target directions, ten runs.

Policy	Method	Accuracy	Macro-F1	Recording BA
Final	IN+Mixup	94.95 ± 1.05	94.72 ± 0.90	99.17 ± 1.76
Final	Unedited bank	95.61 ± 0.72	95.32 ± 0.65	99.58 ± 1.32
Final	Q2	95.72 ± 0.74	95.44 ± 0.63	99.17 ± 1.76
Source-selected	IN+Mixup	95.28 ± 0.90	94.89 ± 0.74	97.92 ± 2.20
Source-selected	Unedited bank	94.78 ± 1.54	94.52 ± 1.25	98.33 ± 2.15
Source-selected	Q2	95.22 ± 1.11	94.85 ± 0.91	98.75 ± 2.01

The single-source extension trains from one JNU speed and evaluates on each of the other two, yielding six directions. Source event analysis uses two legal training contexts from the same source speed; it does not introduce an additional observed condition. Each of the three methods trains three source models per run, and each frozen model supplies two target evaluations.

The fixed-final-state accuracy decreases from the near-ceiling multi-source values to 94.95% for IN+Mixup, 95.61% for the unedited bank and 95.72% for Q2 (Table 11). The 0.77-point Q2 increment over IN+Mixup is mostly accompanied by the change to bank sampling: the increment over the unedited bank is only 0.11 points. Under source selection, Q2 is also slightly below IN+Mixup. These results demonstrate that the operator remains usable with a source-local analyzer, while providing limited evidence for a distinct single-source editing advantage. The six directions are all retained in the appendix.

5.8. Editing measured source signals

Figure 6 shows actual source queries and their event edits. Selection uses metadata alone: for each dataset, take the first task in lexical order at seed 0, class 1, the first source-domain/record identifier, and the middle of its three registered query locations. The corresponding class-0 examples and identifiers are provided in Appendix D. All event coefficients and source contexts are reused from the frozen source bank construction. No target predictions determine these selections.

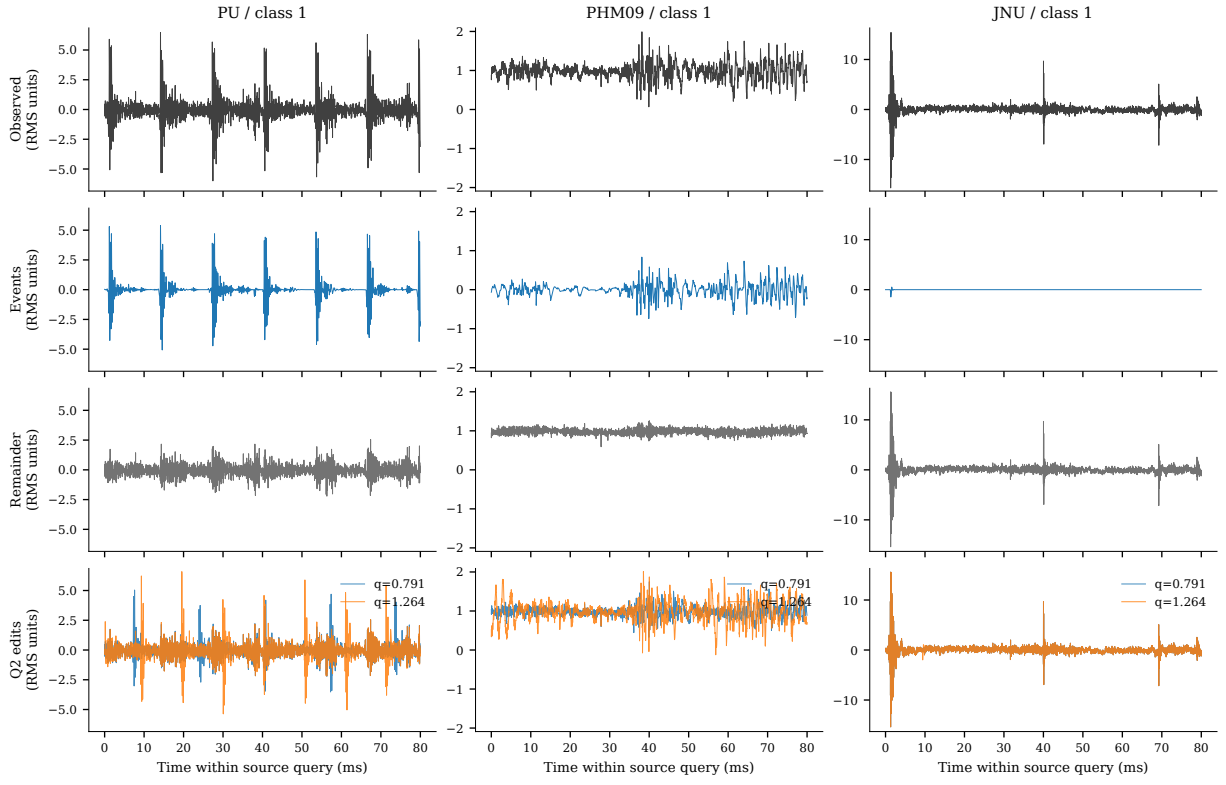


Figure 6: Measured-source examples selected by the fixed metadata rule. Rows show the observed query, estimated event component, retained remainder and two Q2 edits. Columns correspond to PU outer-race class, PHM09 configuration 2 and JNU inner-race class. Time covers the entire registered query; all four rows in a column share the same amplitude scale, divided by that observed query's RMS. The weak estimated event in the selected JNU query is retained in the figure.

The examples show that the editor adapts its effective action to the estimated event component. In the selected PU query, clear repeated transients enter the event component and their locations change. The PHM09 example also retains a substantial observation term. In the JNU example, most observed structure remains in the remainder, so the edit is weak on this query. Neither the event energy nor the remainder energy is interpreted as a separately measured physical-factor share. Classification utility is assessed by the matched experiments rather than by visually selecting a clean decomposition.

6. Discussion

6.1. Interpretation of the mechanism

The proposed editor makes an explicit distinction between event occurrence, effective impulse strength, response shape and the retained observation. Its intervention has a defined action before the diagnostic representation is learned. The unit-rate identity and fixed-response construction establish properties of the operator, while the matched ablations establish its empirical diagnostic contribution. Reconstruction quality alone would not establish either label preservation or improved transfer.

This interpretation also clarifies the role of the other components. IN and Mixup are effective parts of the classifier recipe, and the log-spectrum frontend supplies a compact frequency representation. The matched comparisons keep these components fixed, so the remaining Q2 increment can be attributed to the specified source construction within that recipe. They do not show that the event editor is the largest contributor to the complete system or that it is indispensable on every dataset. The PHM09 alignment baseline and impulse-only control provide substantive alternative explanations for parts of the performance.

The event model specifies a local, model-based intervention. Its estimated coefficients are not direct physical measurements of every impact, and the remainder can contain both fault information and condition effects. The fixed strength–rate coupling is an augmentation rule, not a universal load–speed law. Because the remainder retains its observed coordinates, the edit is narrower than a complete physical change of shaft speed. This selective construction avoids deleting unexplained fault evidence while creating controlled variation in a plausible signal-generation mechanism.

6.2. Deployment and computational implications

The event analyzer and source bank are used only during preparation and training. Q2 and its matched IN+Mixup control have identical deployed networks: 2,200,627 parameters on PU, 2,201,272 on PHM09 and 2,200,756 on JNU. Convolution and linear layers require approximately 5.751 million multiply–accumulates per window, excluding the FFT frontend, normalization, activations and memory operations. Thus the editor adds no deployed trainable parameters or convolution/linear operations.

Source preparation has a real cost. A previously audited three-run preparation campaign built 81 task/run banks from 3372 context groups in 52.69 minutes, including all five rendered rates, control features, I/O and fingerprint checks. This measurement reuses existing 500-iteration numerical fits and is neither the initial inverse-fitting time nor the total ten-run cost. The matched classification arms share 2000 updates and the same batch layout. The practical benefit is therefore a separation between offline source construction and ordinary online diagnosis, rather than a claim of cost-free augmentation.

6.3. Application scope

The experiments evaluate transfer across recorded conditions. PU repeats bearing identities across conditions; PHM09 configuration labels do not identify gearbox serial units; JNU uses separated time blocks of existing recordings. The three datasets therefore support cross-condition diagnosis under the stated protocols, with the multi-source and single-source settings kept distinct. Cross-machine confirmation would require a corresponding evaluation design.

The current evidence is strongest for a selective source intervention that improves a fixed, strong classifier recipe on PU and PHM09. It also shows more repeatable PHM09 training and lower PU recording-level error rates. Deep CORAL remains a strong competitor, source selection changes some rankings, and JNU offers limited multi-source headroom. Future work can examine event-model adequacy and intervention-strength selection using source evidence, together with testing under explicitly measured disturbance and machine changes. Those extensions are separate from the completed experiments reported here.

7. Conclusion

This paper proposes observation-anchored event-mechanism editing for fault diagnosis under unseen operating conditions. A source-only sparse analyzer supplies events and response kernels; the editor changes event timing and effective impulse strength while preserving the remainder of each measured observation. The resulting source views train a conventional classifier with vibration-only deployment.

Across 27 multi-source tasks and ten runs, Q2 improves dataset-equal accuracy by 0.80 percentage points over matched IN+Mixup and by 0.79 points over an unedited-bank control. Timing-only and moment-matched comparisons support a contribution from the specified event construction. The method also reduces PU recording-level missed faults and improves PHM09 repeatability. Strong alignment baselines and the single-source extension show that these benefits are condition-dependent. The results establish observation-anchored event editing as a practical source-training mechanism whose diagnostic contribution can be measured without adding complexity to the deployed model.

A. Task inventory and implementation details

PU uses BCD→A, ACD→B, ABD→C and ABC→D. JNU uses BC→A, AC→B and AB→C. PHM09 adjacent-domain tasks are AC→B, BC→A, BD→C, CE→D, DF→E, EG→F, FH→G, GI→H, HI→J and HJ→I. Its distant-domain tasks are AB→F/G/H/I/J and IJ→A/B/C/D/E, where each target letter defines a separate task.

Source context length is 32768 samples and response support is 4 ms. The four carrier candidates per context are selected from source spectra above four times the largest paired source shaft rate and below 0.44 times the sampling frequency. Candidate peaks are separated by at least $\max(100, 3f_s/N)$ Hz. Each carrier is combined with decay constants 0.25, 0.5 and 1 ms and phases $0, \pi/2, \pi$ and $3\pi/2$. The kernel is an exponentially decaying sinusoid multiplied by a squared cosine taper and normalized by its discrete ℓ_2 norm. Numerical coefficients are obtained by the fixed accelerated proximal solver; the classifier does not update them. The reused numerical fits use 500 proximal iterations. Their penalties are specified from each normalized source context. Let

$$\nu_j = \max\left\{ \frac{\operatorname{median}_{f\in[0.45,0.5]} |\,\mathrm{rFFT}(\bar{x}_j)[f]|}{\sqrt{N\log 2}}, 0.005\sqrt{\frac{1}{N}\sum_t \bar{x}_j[t]^2} \right\}, \tag{15}$$

where f is normalized frequency, and define

$$b(G) = \frac{\nu_1+\nu_2}{2}\sqrt{4+4\sqrt{\log(\max(G,2))}+2\log(\max(G,2))}. \tag{16}$$

Then $\lambda_{s,j} = \nu_j\sqrt{2\log(2\cdot 48N)}$, $\lambda_b = b(4)$ and $\lambda_c = b(\max(G_F,1))$, where G_F is the number of stationary frequency candidates. The proximal step uses the fixed spectral-norm upper bound formed from the convolution, shaft-harmonic and stationary dictionaries. These numerical settings are source-derived and are reused for all edited views of a context.

A.1. Source-domain alignment baselines

For DANN, a source-domain head maps the shared 256-dimensional representation to 128 hidden units with ReLU and then to source-domain logits. The gradient-reversal coefficient follows $2/(1+\exp(-10p))-1$, where p is training progress, and is applied once. Deep CORAL averages covariance discrepancies between each source pair, using unbiased sample covariance and squared Frobenius distance divided by $4d^2$, where $d = 256$. Its alignment weight is one. Both baselines retain the same source pools and fixed training schedule as the ERM baseline. They do not perform target adaptation.

A.2. Exposure controls

Conditional on the original edit mask, half exposure retains an edit with probability 0.5 and otherwise uses the same clean bank query. The expected edited fraction of auxiliary positions falls from 0.5 to 0.25. The expected-energy control instead scales every log-feature edit difference by $\sqrt{0.5}$. All other training choices are unchanged. Standard

Table 12
Exposure controls: window accuracy (%) over ten runs at the fixed final state.

Method	PU	PHM09	JNU
Expected-energy shrinkage	82.51 ± 1.17	71.81 ± 1.24	99.65 ± 0.12
Half-exposure Q2	82.82 ± 1.06	71.94 ± 1.13	99.69 ± 0.08
Standard Q2	83.12 ± 0.94	71.98 ± 0.95	99.67 ± 0.17

Q2 exceeds half exposure by 0.107 points in the primary dataset-equal mean, with only five of ten paired differences positive. Under source selection, half exposure is 0.132 points higher. This sensitivity does not justify replacing the fixed main recipe; both policies and all exposure rows are retained.

B. Complete comparison and paired summaries

The following tables retain all 22 multi-source recipes. The main comparisons use the fixed final state; source selection is reported separately. The poor MOCD equation-literal result is kept as an implementation-interpretation result and is not used to characterize the original method's published performance. Each table reports ten-run mean±sample SD with the task weighting in Eq. (14).

Table 13
Fixed-final-state window accuracy (%) for all 22 complete recipes, ten runs.

Recipe	PU	PHM09	JNU
Author-backbone CE	80.63 ± 1.38	51.37 ± 1.93	83.95 ± 7.88
Author-backbone IN+Mixup	81.85 ± 0.85	68.23 ± 1.06	95.31 ± 4.86
Archived identity+Mixup	80.86 ± 0.39	67.68 ± 1.40	99.68 ± 0.26
Archived IN+Mixup	81.35 ± 0.94	71.35 ± 1.39	99.59 ± 0.14
CCDG (author feature core)	81.13 ± 1.16	57.47 ± 1.73	87.19 ± 8.11
Earlier event editor	82.59 ± 0.97	71.46 ± 1.35	99.55 ± 0.17
MOCD (literal equation)	80.66 ± 1.76	10.31 ± 1.47	29.66 ± 24.80
MOCD (prose interpretation)	80.50 ± 0.93	65.26 ± 2.51	95.14 ± 2.86
Whole-waveform scaling	80.63 ± 1.29	68.34 ± 1.02	99.11 ± 0.30
Archived intact control	81.13 ± 0.82	71.60 ± 1.44	99.60 ± 0.21
DANN (source domains)	80.81 ± 1.52	52.94 ± 2.40	82.47 ± 10.02
Deep CORAL (source domains)	82.99 ± 1.14	72.65 ± 1.25	99.87 ± 0.06
ERM (shared CNN+IN)	82.97 ± 1.34	63.81 ± 0.99	99.71 ± 0.14
Impulse strength only	81.46 ± 1.19	72.58 ± 1.52	99.66 ± 0.12
Matched IN+Mixup	81.44 ± 0.98	71.30 ± 1.85	99.64 ± 0.08
Timing, endpoint rates	82.82 ± 1.02	71.59 ± 1.16	99.62 ± 0.19
Unedited bank	81.47 ± 1.09	71.29 ± 1.90	99.64 ± 0.13
Global moment matching	81.36 ± 0.87	71.69 ± 1.27	99.69 ± 0.07
Proposed Q2	83.12 ± 0.94	71.98 ± 0.95	99.67 ± 0.17
Timing, Q2 rates	82.84 ± 0.85	71.26 ± 0.84	99.63 ± 0.13
Expected-energy shrinkage	82.51 ± 1.17	71.81 ± 1.24	99.65 ± 0.12
Half-exposure Q2	82.82 ± 1.06	71.94 ± 1.13	99.69 ± 0.08

Table 14
Source-selected window accuracy (%) for all 22 complete recipes, ten runs.

Recipe	PU	PHM09	JNU
Author-backbone CE	80.63 ± 1.38	50.97 ± 2.17	83.94 ± 7.42
Author-backbone IN+Mixup	81.85 ± 0.85	65.20 ± 1.40	95.03 ± 5.11
Archived identity+Mixup	80.89 ± 0.36	67.05 ± 2.00	99.04 ± 1.04
Archived IN+Mixup	81.42 ± 0.91	71.82 ± 1.52	99.69 ± 0.17
CCDG (author feature core)	81.13 ± 1.16	55.76 ± 1.26	86.71 ± 8.00
Earlier event editor	82.58 ± 0.99	71.57 ± 1.37	99.69 ± 0.17
MOCD (literal equation)	80.80 ± 1.56	11.82 ± 0.79	53.50 ± 28.66
MOCD (prose interpretation)	80.52 ± 1.36	60.79 ± 3.54	95.70 ± 4.06
Whole-waveform scaling	81.21 ± 1.17	67.48 ± 0.84	99.25 ± 0.30
Archived intact control	81.15 ± 0.78	71.68 ± 1.19	99.71 ± 0.11
DANN (source domains)	80.85 ± 2.27	52.49 ± 2.30	79.41 ± 9.55
Deep CORAL (source domains)	83.04 ± 1.04	71.97 ± 1.14	99.88 ± 0.06
ERM (shared CNN+IN)	82.95 ± 1.02	63.35 ± 0.93	99.67 ± 0.16
Impulse strength only	81.53 ± 1.08	72.95 ± 0.92	99.69 ± 0.09
Matched IN+Mixup	81.38 ± 0.92	71.40 ± 1.50	99.72 ± 0.12
Timing, endpoint rates	82.68 ± 1.01	71.79 ± 1.07	99.70 ± 0.10
Unedited bank	81.49 ± 1.16	71.39 ± 1.46	99.71 ± 0.11
Global moment matching	81.59 ± 1.35	73.00 ± 1.55	99.75 ± 0.09
Proposed Q2	83.00 ± 1.05	71.58 ± 1.22	99.74 ± 0.06
Timing, Q2 rates	82.82 ± 0.86	70.92 ± 0.93	99.69 ± 0.12
Expected-energy shrinkage	82.49 ± 1.37	71.97 ± 1.49	99.68 ± 0.10
Half-exposure Q2	82.84 ± 1.27	72.18 ± 1.17	99.71 ± 0.07

Table 15
Dataset-equal paired accuracy differences: Q2 minus each control (percentage points). Intervals are descriptive 95% t_9 intervals over ten paired run averages.

Control	Mean difference	95% interval	Positive runs
Matched IN+Mixup	+0.801	[+0.342, +1.259]	9/10
Unedited bank	+0.791	[+0.312, +1.270]	9/10
Timing, endpoint rates	+0.247	[−0.021, +0.514]	7/10
Timing, Q2 rates	+0.346	[+0.243, +0.448]	10/10
Impulse only	+0.355	[−0.013, +0.724]	9/10
Global moment matching	+0.676	[+0.325, +1.027]	9/10
Half-exposure Q2	+0.107	[−0.163, +0.378]	5/10
Expected-energy shrinkage	+0.268	[+0.081, +0.456]	9/10
Deep CORAL	−0.247	[−0.597, +0.103]	3/10
ERM	+2.759	[+2.406, +3.112]	10/10

These post-result intervals are unadjusted for multiple comparisons and conditional on the fixed task panel; they are not intervals over independent machine populations.

Table 16
Fixed-final-state window macro-F1 (%) for all 22 recipes, ten runs.

Recipe	PU	PHM09	JNU
Author-backbone CE	78.68 ± 1.18	44.91 ± 2.13	85.24 ± 6.36
Author-backbone IN+Mixup	80.32 ± 0.86	64.50 ± 1.49	96.14 ± 3.66
Archived identity+Mixup	79.81 ± 0.49	63.98 ± 1.43	99.67 ± 0.26
Archived IN+Mixup	79.92 ± 0.90	67.85 ± 1.76	99.57 ± 0.15
CCDG (author feature core)	79.11 ± 1.10	52.04 ± 1.86	89.29 ± 6.70
Earlier event editor	81.01 ± 0.93	68.36 ± 1.29	99.53 ± 0.18
MOCD (literal equation)	78.11 ± 1.78	3.56 ± 0.66	26.02 ± 23.92
MOCD (prose interpretation)	77.90 ± 0.94	60.41 ± 2.84	94.51 ± 2.02
Whole-waveform scaling	79.03 ± 1.24	64.61 ± 1.05	99.03 ± 0.31
Archived intact control	79.64 ± 0.90	68.17 ± 1.82	99.57 ± 0.21
DANN (source domains)	79.54 ± 1.31	46.15 ± 2.75	79.61 ± 9.72
Deep CORAL (source domains)	81.27 ± 1.00	68.69 ± 1.44	99.84 ± 0.07
ERM (shared CNN+IN)	81.51 ± 1.34	58.51 ± 1.21	99.67 ± 0.17
Impulse strength only	80.03 ± 1.09	69.63 ± 1.63	99.64 ± 0.12
Matched IN+Mixup	80.07 ± 1.02	67.75 ± 2.15	99.62 ± 0.09
Timing, endpoint rates	81.23 ± 0.87	68.43 ± 1.06	99.60 ± 0.21
Unedited bank	80.00 ± 0.96	67.81 ± 2.06	99.63 ± 0.13
Global moment matching	79.39 ± 0.69	68.59 ± 1.38	99.68 ± 0.07
Proposed Q2	81.57 ± 0.85	68.66 ± 1.18	99.65 ± 0.17
Timing, Q2 rates	81.33 ± 0.75	67.87 ± 0.95	99.61 ± 0.14
Expected-energy shrinkage	80.97 ± 1.02	68.61 ± 1.35	99.62 ± 0.11
Half-exposure Q2	81.32 ± 0.97	68.74 ± 1.33	99.67 ± 0.09

Table 17
Source-selected window macro-F1 (%) for all 22 recipes, ten runs.

Recipe	PU	PHM09	JNU
Author-backbone CE	78.68 ± 1.18	44.45 ± 2.46	84.73 ± 5.80
Author-backbone IN+Mixup	80.32 ± 0.86	61.21 ± 1.66	95.73 ± 3.77
Archived identity+Mixup	79.83 ± 0.42	63.43 ± 1.98	99.05 ± 0.98
Archived IN+Mixup	79.94 ± 0.79	68.09 ± 1.77	99.68 ± 0.17
CCDG (author feature core)	79.11 ± 1.10	50.04 ± 1.30	88.39 ± 6.80
Earlier event editor	81.01 ± 0.97	68.28 ± 1.36	99.67 ± 0.18
MOCD (literal equation)	78.34 ± 1.74	2.82 ± 0.29	47.41 ± 27.54
MOCD (prose interpretation)	77.87 ± 1.50	55.26 ± 4.02	95.59 ± 3.10
Whole-waveform scaling	79.53 ± 1.07	63.38 ± 0.88	99.20 ± 0.31
Archived intact control	79.68 ± 0.85	68.01 ± 1.33	99.69 ± 0.11
DANN (source domains)	79.41 ± 1.92	45.56 ± 2.91	76.58 ± 8.47
Deep CORAL (source domains)	81.32 ± 0.81	68.00 ± 1.31	99.86 ± 0.08
ERM (shared CNN+IN)	81.39 ± 1.04	57.94 ± 1.15	99.62 ± 0.20
Impulse strength only	80.07 ± 1.04	69.64 ± 1.09	99.67 ± 0.09
Matched IN+Mixup	80.01 ± 0.97	67.68 ± 1.77	99.71 ± 0.12
Timing, endpoint rates	81.11 ± 0.82	68.45 ± 1.17	99.68 ± 0.10
Unedited bank	79.99 ± 1.06	67.64 ± 1.60	99.69 ± 0.10
Global moment matching	79.55 ± 1.21	69.64 ± 1.78	99.74 ± 0.09
Proposed Q2	81.44 ± 0.93	68.06 ± 1.54	99.73 ± 0.06
Timing, Q2 rates	81.28 ± 0.76	67.24 ± 1.24	99.67 ± 0.12
Expected-energy shrinkage	80.92 ± 1.23	68.56 ± 1.68	99.67 ± 0.10
Half-exposure Q2	81.31 ± 1.18	68.84 ± 1.30	99.70 ± 0.07

Table 18
Fixed-final-state recording balanced accuracy (%) for all 22 recipes, ten runs.

Recipe	PU	PHM09	JNU
Author-backbone CE	82.47 ± 1.54	53.41 ± 2.18	92.50 ± 4.73
Author-backbone IN+Mixup	86.70 ± 1.02	72.50 ± 2.02	98.33 ± 3.51
Archived identity+Mixup	86.20 ± 0.87	70.25 ± 2.02	100.00 ± 0.00
Archived IN+Mixup	86.57 ± 1.47	74.62 ± 2.25	100.00 ± 0.00
CCDG (author feature core)	83.53 ± 2.15	59.00 ± 2.16	94.17 ± 4.03
Earlier event editor	87.42 ± 1.01	76.97 ± 1.52	100.00 ± 0.00
MOCD (literal equation)	84.84 ± 3.23	11.09 ± 1.57	30.00 ± 25.22
MOCD (prose interpretation)	86.01 ± 1.98	67.72 ± 3.46	100.00 ± 0.00
Whole-waveform scaling	86.69 ± 1.54	72.88 ± 2.20	100.00 ± 0.00
Archived intact control	86.51 ± 1.46	75.69 ± 2.40	100.00 ± 0.00
DANN (source domains)	87.01 ± 1.96	54.44 ± 1.71	88.33 ± 8.05
Deep CORAL (source domains)	86.86 ± 1.92	75.56 ± 1.52	100.00 ± 0.00
ERM (shared CNN+IN)	86.97 ± 2.43	66.16 ± 1.02	100.00 ± 0.00
Impulse strength only	86.86 ± 1.12	77.44 ± 2.19	100.00 ± 0.00
Matched IN+Mixup	86.58 ± 1.24	74.94 ± 2.95	100.00 ± 0.00
Timing, endpoint rates	87.79 ± 1.11	76.91 ± 1.64	100.00 ± 0.00
Unedited bank	86.92 ± 1.13	75.47 ± 2.85	100.00 ± 0.00
Global moment matching	87.06 ± 0.99	75.97 ± 2.15	100.00 ± 0.00
Proposed Q2	87.81 ± 0.83	76.72 ± 1.71	100.00 ± 0.00
Timing, Q2 rates	87.71 ± 0.79	76.06 ± 0.97	100.00 ± 0.00
Expected-energy shrinkage	87.81 ± 0.95	75.69 ± 1.59	100.00 ± 0.00
Half-exposure Q2	87.68 ± 1.09	75.53 ± 1.93	100.00 ± 0.00

Table 19
Source-selected recording balanced accuracy (%) for all 22 recipes, ten runs.

Recipe	PU	PHM09	JNU
Author-backbone CE	82.47 ± 1.54	53.25 ± 2.16	92.50 ± 4.73
Author-backbone IN+Mixup	86.70 ± 1.02	69.19 ± 2.30	97.50 ± 4.03
Archived identity+Mixup	86.27 ± 0.74	69.16 ± 2.64	100.00 ± 0.00
Archived IN+Mixup	86.40 ± 1.20	75.22 ± 2.13	100.00 ± 0.00
CCDG (author feature core)	83.53 ± 2.15	57.72 ± 1.74	94.17 ± 4.03
Earlier event editor	87.27 ± 1.16	76.47 ± 1.65	100.00 ± 0.00
MOCD (literal equation)	84.60 ± 3.70	11.88 ± 0.92	52.50 ± 25.17
MOCD (prose interpretation)	85.67 ± 1.32	62.38 ± 4.22	100.00 ± 0.00
Whole-waveform scaling	86.99 ± 1.17	71.66 ± 1.60	100.00 ± 0.00
Archived intact control	86.46 ± 1.59	75.25 ± 1.34	100.00 ± 0.00
DANN (source domains)	86.97 ± 2.30	53.91 ± 2.32	87.50 ± 7.08
Deep CORAL (source domains)	86.77 ± 1.66	75.25 ± 1.36	100.00 ± 0.00
ERM (shared CNN+IN)	86.73 ± 1.63	65.47 ± 0.85	100.00 ± 0.00
Impulse strength only	86.88 ± 1.12	77.16 ± 1.62	100.00 ± 0.00
Matched IN+Mixup	86.73 ± 1.15	74.38 ± 1.54	100.00 ± 0.00
Timing, endpoint rates	87.65 ± 1.02	76.16 ± 1.79	100.00 ± 0.00
Unedited bank	86.85 ± 1.10	74.78 ± 2.52	100.00 ± 0.00
Global moment matching	87.14 ± 1.15	76.81 ± 2.24	100.00 ± 0.00
Proposed Q2	87.71 ± 0.76	75.22 ± 1.58	100.00 ± 0.00
Timing, Q2 rates	87.59 ± 0.75	75.03 ± 0.80	100.00 ± 0.00
Expected-energy shrinkage	87.67 ± 1.00	75.97 ± 2.20	100.00 ± 0.00
Half-exposure Q2	87.50 ± 1.07	76.41 ± 1.99	100.00 ± 0.00

C. Task-level and extension results

Table 20
All 27 multi-source tasks: fixed-final-state window accuracy (%), ten runs.

Dataset	Sources→target	IN+Mixup	Q2	Deep CORAL
PU	ABC→D	38.43 ± 4.23	44.84 ± 3.46	46.93 ± 4.65
PU	ABD→C	98.42 ± 0.29	98.31 ± 0.25	98.40 ± 0.30
PU	ACD→B	91.94 ± 1.45	92.32 ± 1.66	90.27 ± 2.58
PU	BCD→A	96.95 ± 0.31	97.01 ± 0.24	96.36 ± 0.51
PHM09	AC→B (adj.)	96.86 ± 2.04	93.24 ± 2.03	96.43 ± 1.78
PHM09	BC→A (adj.)	98.02 ± 0.95	97.76 ± 1.25	95.22 ± 2.45
PHM09	BD→C (adj.)	78.89 ± 7.13	80.57 ± 5.63	81.41 ± 6.41
PHM09	CE→D (adj.)	91.52 ± 3.71	87.55 ± 3.27	79.07 ± 4.48
PHM09	DF→E (adj.)	72.94 ± 3.87	75.32 ± 3.99	74.59 ± 2.91
PHM09	EG→F (adj.)	81.91 ± 3.65	75.92 ± 5.84	81.58 ± 3.24
PHM09	FH→G (adj.)	57.33 ± 5.28	64.75 ± 6.34	63.06 ± 2.94
PHM09	GI→H (adj.)	82.15 ± 4.71	82.13 ± 5.23	78.54 ± 3.47
PHM09	HI→J (adj.)	89.04 ± 3.72	86.84 ± 3.41	86.68 ± 4.43
PHM09	HJ→I (adj.)	52.81 ± 4.89	57.27 ± 4.46	61.45 ± 4.17
PHM09	AB→F (dist.)	69.69 ± 6.10	74.29 ± 3.18	72.07 ± 6.47
PHM09	AB→G (dist.)	52.92 ± 2.84	63.25 ± 3.38	54.95 ± 4.70
PHM09	AB→H (dist.)	51.68 ± 9.85	53.17 ± 7.40	52.95 ± 10.04
PHM09	AB→I (dist.)	47.03 ± 5.53	45.25 ± 4.91	52.95 ± 2.73
PHM09	AB→J (dist.)	36.05 ± 3.33	36.62 ± 2.43	37.47 ± 0.26
PHM09	IJ→A (dist.)	49.76 ± 10.83	49.78 ± 7.23	53.57 ± 10.10
PHM09	IJ→B (dist.)	77.11 ± 4.25	68.28 ± 7.66	64.52 ± 12.19
PHM09	IJ→C (dist.)	77.99 ± 8.68	78.81 ± 3.67	79.43 ± 7.81
PHM09	IJ→D (dist.)	77.59 ± 7.26	84.84 ± 6.24	88.93 ± 4.21
PHM09	IJ→E (dist.)	84.62 ± 6.05	84.02 ± 5.83	98.24 ± 0.82
JNU	BC→A	99.18 ± 0.21	99.24 ± 0.49	99.65 ± 0.16
JNU	AC→B	99.76 ± 0.19	99.79 ± 0.10	99.96 ± 0.05
JNU	AB→C	99.97 ± 0.03	99.97 ± 0.03	99.99 ± 0.03

C.1. JNU single-source extension

The same temporal source roles and classifier schedule are retained. Numerical analysis uses source-local contexts. No target speed or second-source recording is introduced. The six directions below evaluate the same three trained source models per method/run; the two target evaluations of one source model are not counted as two independently trained models.

Table 21
All JNU single-source directions: fixed-final-state window accuracy (%), ten runs.

Direction	IN+Mixup	Unedited bank	Q2
A→B	94.14 ± 2.89	95.92 ± 1.52	95.92 ± 1.18
A→C	91.95 ± 4.27	92.52 ± 3.84	93.11 ± 4.00
B→A	98.62 ± 0.58	98.94 ± 0.47	98.94 ± 0.45
B→C	99.97 ± 0.03	99.88 ± 0.11	99.91 ± 0.08
C→A	88.36 ± 1.51	89.19 ± 1.61	89.12 ± 1.56
C→B	96.65 ± 0.54	97.18 ± 0.41	97.31 ± 0.37

Table 22
PU recording-level comparison, ten runs, source-selected state. All scores are percentages.

Recipe	Accuracy	Macro-F1	Balanced accuracy
Q2, short vibration input	84.74 ± 1.07	83.90 ± 1.07	87.71 ± 0.76
Matched IN+Mixup	83.58 ± 1.20	82.99 ± 1.40	86.73 ± 1.15
Unedited bank	83.83 ± 1.26	83.04 ± 1.16	86.85 ± 1.10
Native 40-rev. author CE	85.73 ± 1.64	82.15 ± 1.45	84.55 ± 2.05
Native 40-rev. CE+real healthy	85.28 ± 1.92	82.37 ± 1.76	85.74 ± 2.13
Native 40-rev. PCR	82.47 ± 0.71	80.26 ± 1.02	84.35 ± 1.14
Short-input author CE	83.96 ± 0.75	76.05 ± 0.62	75.66 ± 0.94
Short-input CE+real healthy	83.90 ± 1.23	75.90 ± 1.19	75.79 ± 1.26
Short-input PCR	83.64 ± 1.02	75.55 ± 1.20	74.84 ± 1.53

Native PCR uses recorded speed to construct 40-revolution inputs. The comparison aligns recording roles, not input windows or architectures. This is a declared cross-condition adaptation of a single-source cross-machine method.

D. Additional measured-source illustrations

Class-0 source examples use the same fixed metadata selection rule as Fig. 6. Class 0 is healthy for PU and JNU; for PHM09 it is configuration 1. The source examples are illustrations of operator action, not examples selected for target accuracy. Their identifiers and edit energies are provided below.

Table 23
Fixed source examples. Each RMS ratio uses the observed query RMS as denominator; edit RMS measures the difference from that observation. These are waveform ratios, not shares of a physical factor.

Dataset	Class	Event RMS ratio	Low-q edit ratio	High-q edit ratio
PU	0	0.0644	0.0725	0.0986
PU	1	0.7966	0.9404	1.0730
PHM09	0	0.1402	0.1650	0.2361
PHM09	1	0.1729	0.1981	0.2958
JNU	0	0.1200	0.1197	0.1628
JNU	1	0.0464	0.0464	0.0712

Table 24
Identifiers of the six measured source examples. Query starts are zero-based sample indices in the source recording.

Dataset	Class	Source / recording	Query start
PU	0	A/K001/1	79360
PU	1	A/KA01/1	79360
PHM09	0	A/spur 1_30hz_High_1.txt	79253
PHM09	1	A/spur 2_30hz_High_1.txt	79253
JNU	0	B/n800_3_2.csv	79920
JNU	1	B/ib800_2.csv	79920

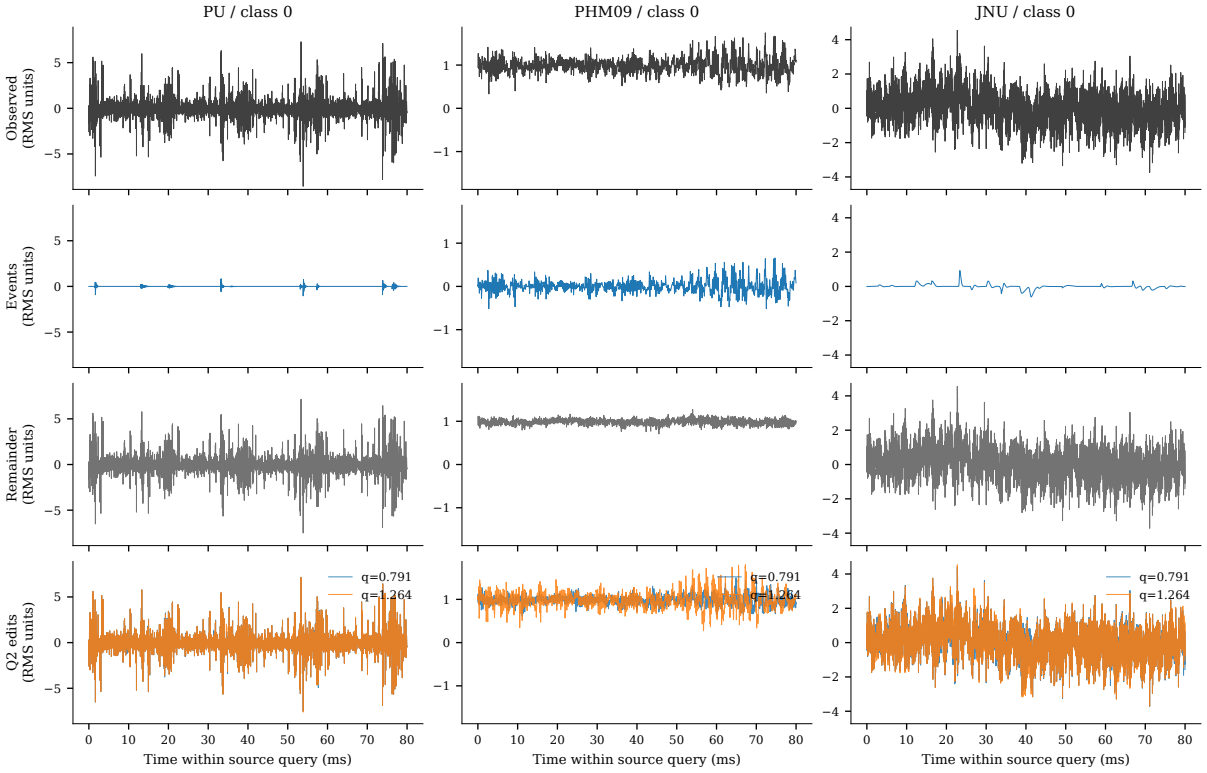


Figure 7: Fixed class-0 source examples. PHM09 configuration 1 is not assumed to be a verified healthy reference. The within-column amplitude convention is the same as in Fig. 6.

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